DATA ANALYST PORTFOLIO PROJECT · THE STORY BEHIND IT

The Data I Had, But Didn't Know How to Use

Sun Sahar · Companion to the presentation and the SQL file

The presentation shows the findings. This page gives the context behind them: the business, what I did at the time, and what changed in how I think.

1. The business

I ran a small trading-education business. A free course, promoted with Meta ads, brought people in; we tried to reach each lead by phone to offer a paid course with personal mentorship. People became paying customers through several routes: word of mouth, ads, webinars and personal calls.

I ran the advertising myself from mid-2024 to early 2026. In 2024 we spent little while we learned what to offer. From February to August 2025 we ran free-course lead campaigns; March was the busiest month, and calling the new leads kept us fully occupied. In late 2025 I tested new offers: a ₪97 mini-course, webinars and messaging campaigns. Throughout, I mainly judged results by two numbers: how many leads came in, and what each one cost. Getting from a lead to a sale was the hardest part of the business, and the part I measured least.

2. Why I went back to the data

As a Data Analyst student I wanted a real project, not a tutorial dataset. My own business gave me something rare: I could compare what the data said with what I had actually done and believed. The question I set myself was: *what could this data have told me then, and what could it never tell me?*

3. How the data changed my understanding

> - **I was looking at the wrong step.** When costs rose I suspected the form and the offer. Most of the change happened before anyone clicked.
> - **The step that paid was the one I measured least.** Meta counted every lead automatically. Calls, meetings and sales were barely recorded.
> - **I never recorded where my paying customers came from.** Payments were higher while the lead campaigns ran, but sales were happening before the ads too. Without a channel on each customer, I cannot say which ones the ads brought.
> - **Some of my memories did not match the data.** I remembered every campaign as a lead campaign; almost a quarter of the budget went to exposure and offer tests.

4. How I worked

I checked the data before trusting it: the key reported totals were cross-checked between Python and BigQuery before any analysis began, and I kept my own recollections in a separate document from the data. During validation, I found a discrepancy between an earlier business record indicating 46 students and the payment dataset containing 38 distinct paying customers across 50 payments (₪142,234, excluding VAT). Since the two figures could not be reliably reconciled, I used the 38 customers supported by the payment records for the financial analysis. Customer names were replaced by anonymous IDs, and no payment is linked to an ad, so the project reports no acquisition cost, return on ad spend or profit. The full step-by-step method is in the project's methodology page.

5. What I would measure differently

- Tag every lead and every paying customer with its source: ad, webinar, referral or call.
- Record each lead's journey: called, meeting scheduled, meeting held, sale, amount.
- Steer by cost per scheduled meeting rather than cost per lead.

6. How this project was built

I built this project with AI tools. I did not write the SQL or Python code from scratch: AI tools wrote it. My part was to supply the business context and the data, to define and refine the questions, to run the queries in BigQuery, to build the dashboard with guidance, to challenge results that did not match what I knew about the business, and to interpret what the findings mean. I am still learning to write this code on my own, and every query is documented so that I can.

Presentation, SQL, case studies and methodology: github.com/Ssahar6/python
All data is my own business's data: aggregate advertising figures and anonymised payments (excl. VAT). No customer names or contact details appear anywhere in the project.